Netflix Business Case Study

```
!gdown https://d2beiqkhq929f0.cloudfront.net/public_assets/assets/000/000/940/original/netflix.csv
```

```
Downloading...
From: https://d2beiqkhq929f0.cloudfront.net/public_assets/assets/000/000/940/original/netflix.csv
To: /content/netflix.csv
100% 3.40M/3.40M [00:00<00:00, 43.7MB/s]
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
import warnings
warnings.filterwarnings('ignore',category = DeprecationWarning)
warnings.filterwarnings('ignore',category = FutureWarning)
warnings.filterwarnings('ignore',category = UserWarning)
warnings.filterwarnings('ignore',category = RuntimeWarning)
```

```
data = pd.read_csv('netflix.csv')
```

```
data.head()
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	descript
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	2021-09-25	2020	PG-13	90 min	Documentaries	As her fa near end c life, filn
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	' cros paths party, a C Tow
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	2021-09-24	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protec family fr pow drug
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	2021-09-24	2021	TV-MA	1 Season	Docuseries, Reality TV	Fe flirtations toilet tal down ar
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a c coac cer knov trai

```
data.shape
```

```
(8807, 14)
```

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 14 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      6173 non-null   object
 4   cast          7982 non-null   object
 5   country       7976 non-null   object
 6   date_added    8709 non-null   datetime64[ns]
```

```
 7   release_year  8807 non-null   int64
 8   rating        8803 non-null   object
 9   duration      8804 non-null   object
 10  listed_in     8807 non-null   object
 11  description   8807 non-null   object
 12  year_added    8709 non-null   float64
 13  month_added   8709 non-null   float64
dtypes: datetime64[ns](1), float64(2), int64(1), object(10)
memory usage: 963.4+ KB
```

1. Problem Statement

Netflix is a global streaming platform with a diverse catalog of movies and TV shows produced across different countries and genres. With increasing competition in the streaming market, Netflix needs to make data-driven decisions on what type of content to produce and which regions to focus on in order to maximize user engagement and grow its subscriber base.

The objective of this analysis is to explore Netflix's existing content library and identify patterns related to content type, release trends, country-wise distribution, genres, and ratings. These insights will help Netflix understand audience preferences and guide strategic decisions regarding future content production and regional expansion.

Basic Metrics Analysis

The dataset contains 8807 rows and 12 columns, representing movies and TV shows available on Netflix.

The content is categorized into two types: Movies and TV Shows.

The dataset spans multiple release years, indicating both older and recent content.

Each title includes metadata such as country of production, rating, genre, duration, and date added to Netflix.

2. Observations on the shape of data, data types of all the attributes, conversion of categorical attributes to 'category' (If required), missing value detection, statistical summary

```
data.shape
```

```
(8807, 14)
```

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
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 4   cast          7982 non-null   object
 5   country       7976 non-null   object
 6   date_added    8709 non-null   datetime64[ns]
 7   release_year  8807 non-null   int64
 8   rating        8803 non-null   object
 9   duration      8804 non-null   object
 10  listed_in     8807 non-null   object
 11  description   8807 non-null   object
 12  year_added    8709 non-null   float64
 13  month_added   8709 non-null   float64
dtypes: datetime64[ns](1), float64(2), int64(1), object(10)
memory usage: 963.4+ KB
```

Convert categorical columns

```
cat_cols = ['type', 'rating', 'listed_in']
data[cat_cols] = data[cat_cols].astype('category')
```

```
data.isnull().sum()
```

	0
show_id	0
type	0
title	0
director	2634
cast	825
country	831
date_added	10
release_year	0
rating	4
duration	3
listed_in	0
description	0

dtype: int64

```
data.describe(include='all')
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in
count	8807	8807	8807	6173	7982	7976	8709	8807.000000	8803	8804	8807
unique	8807	2	8807	4528	7692	748	NaN	NaN	17	220	514
top	s8807	Movie	Zubaan	Rajiv Chilaka	David Attenborough	United States	NaN	NaN	TV-MA	1 Season	Dramas, International Movies
freq	1	6131	1	19	19	2818	NaN	NaN	3207	1793	362
mean	NaN	NaN	NaN	NaN	NaN	NaN	2019-05-23 01:45:29.452290816	2014.180198	NaN	NaN	NaN
min	NaN	NaN	NaN	NaN	NaN	NaN	2008-01-01 00:00:00	1925.000000	NaN	NaN	NaN
25%	NaN	NaN	NaN	NaN	NaN	NaN	2018-04-20 00:00:00	2013.000000	NaN	NaN	NaN
50%	NaN	NaN	NaN	NaN	NaN	NaN	2019-07-12 00:00:00	2017.000000	NaN	NaN	NaN
75%	NaN	NaN	NaN	NaN	NaN	NaN	2020-08-26 00:00:00	2019.000000	NaN	NaN	NaN
max	NaN	NaN	NaN	NaN	NaN	NaN	2021-09-25 00:00:00	2021.000000	NaN	NaN	NaN
std	NaN	NaN	NaN	NaN	NaN	NaN	NaN	8.819312	NaN	NaN	NaN

Observations on Data Shape and Types

The dataset consists of 8807 rows and 12 columns.

The dataset contains a mix of categorical, numerical, date, and text-based attributes.

Columns such as type, rating, and listed_in are categorical in nature.

release_year is a numerical variable representing the year of release.

Text-heavy columns like title, description, cast, and director are stored as object types.

Categorical columns were converted to the category data type for better memory efficiency and analysis.

Missing Value Analysis:

Several columns contain missing values, most notably:

director

cast

country

Missing values are minimal or absent in key attributes such as type, title, and release_year.

The presence of missing values is expected, especially for older or regional content.

Statistical Summary:

The dataset covers content released across a wide range of years.

The release_year variable shows that the dataset includes both old and recent titles.

Categorical variables show high diversity, indicating a wide variety of content types, genres, and countries.

3. Non-Graphical Analysis: Value counts and unique attributes

```
data['type'].value_counts()
```

type	count
Movie	6131
TV Show	2676

dtype: int64

```
data['country'].value_counts().head(10)
```

country	count
United States	2818
India	972
United Kingdom	419
Japan	245
South Korea	199
Canada	181
Spain	145
France	124
Mexico	110
Egypt	106

dtype: int64

```
data['rating'].value_counts()
```

rating	count
TV-MA	3207
TV-14	2160
TV-PG	863
R	799
PG-13	490
TV-Y7	334
TV-Y	307
PG	287
TV-G	220
NR	80
G	41
TV-Y7-FV	6
NC-17	3
UR	3
66 min	1
84 min	1
74 min	1

dtype: int64

```
data['listed_in'].value_counts().head(10)
```

listed_in	count
Dramas, International Movies	362
Documentaries	359
Stand-Up Comedy	334
Comedies, Dramas, International Movies	274
Dramas, Independent Movies, International Movies	252
Kids' TV	220
Children & Family Movies	215
Children & Family Movies, Comedies	201
Documentaries, International Movies	186
Dramas, International Movies, Romantic Movies	180

dtype: int64

```
data.nunique()
```

	0
show_id	8807
type	2
title	8807
director	4528
cast	7692
country	748
date_added	1699
release_year	74
rating	17
duration	220
listed_in	514
description	8775
year_added	14
month_added	12

dtype: int64

Observations

1.Movies > TV Shows (overall)

2.Top countries usually:

United States

India

United Kingdom

Canada

Japan

3.Most frequent ratings:

TV-MA

TV-14

R

Means Netflix targets teen & adult audiences

4.Top 5 Genres

International Movies

Dramas

Comedies

Action & Adventure

Documentaries

5.Unique Attributes: The dataset contains a large number of unique titles, directors, and cast members, indicating a diverse content library with contributions from multiple creators and regions.

4. Visual Analysis - Univariate, Bivariate after pre-processing of the data

Note: Pre-processing involves unnesting of the data in columns like Actor, Director, Country

4.1 For continuous variable(s): Distplot, countplot, histogram for univariate analysis (10 Points)

4.2 For categorical variable(s): Boxplot (10 Points)

Convert date_added to datetime

```
data['date_added'] = pd.to_datetime(data['date_added'], errors='coerce')
```

```
data['year_added'] = data['date_added'].dt.year
data['month_added'] = data['date_added'].dt.month
```

Separate Movie & TV Show duration

Movies → minutes

TV Shows → seasons

They cannot be analyzed together.

```
movies = data[data['type'] == 'Movie'].copy()
```

```
movies['duration_minutes'] = movies['duration'].str.extract(r'(\d+)')
movies['duration_minutes'] = pd.to_numeric(movies['duration_minutes'], errors='coerce')
movies_clean = movies.dropna(subset=['duration_minutes'])
```

We do this before unnesting to avoid errors

```
data[['cast', 'director', 'country']] = data[['cast', 'director', 'country']].fillna('Unknown')
```

Unnest Country

```
country_df = data.copy()
country_df['country'] = country_df['country'].str.split(', ')
country_df = country_df.explode('country')
country_df = country_df.reset_index(drop=True)
```

each country is in a separate row

Unnest Director

```
director_df = data.copy()
director_df['director'] = director_df['director'].str.split(', ')
director_df = director_df.explode('director')
director_df = director_df.reset_index(drop=True)
```

Unnset Cast

```
cast_df = data.copy()
cast_df['cast'] = cast_df['cast'].str.split(', ')
cast_df = cast_df.explode('cast')
cast_df = cast_df.reset_index(drop=True)
```

```
country_df[['title', 'country']].head(10)
```

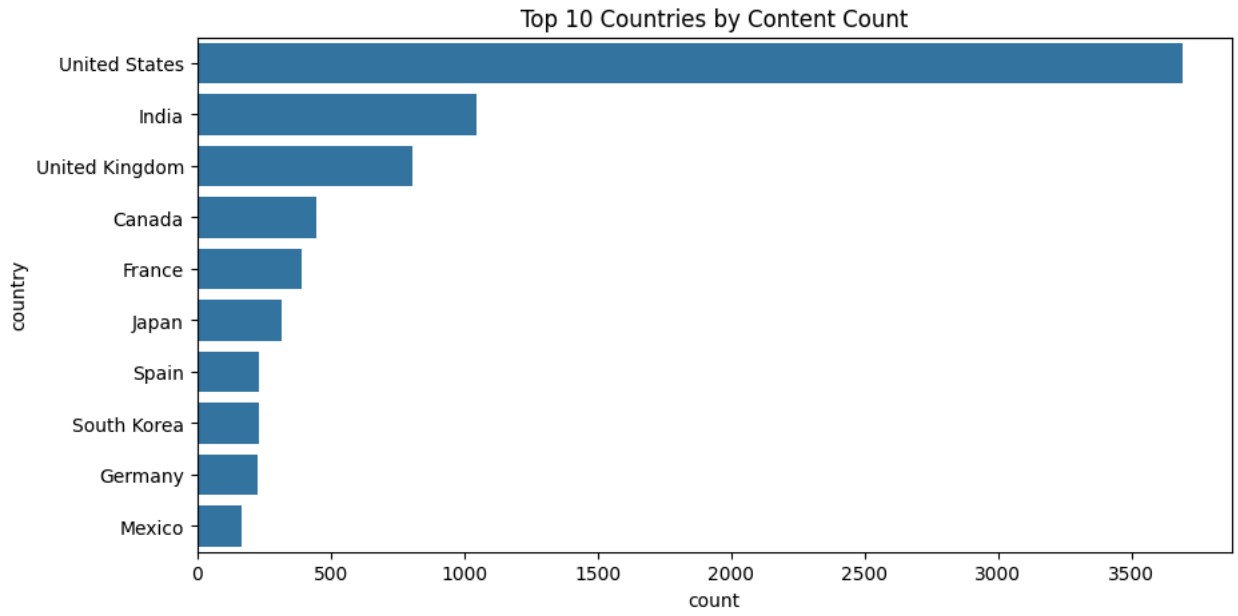
	title	country
0	Dick Johnson Is Dead	United States
1	Blood & Water	South Africa
2	Ganglands	NaN
3	Jailbirds New Orleans	NaN
4	Kota Factory	India
5	Midnight Mass	NaN
6	My Little Pony: A New Generation	NaN
7	Sankofa	United States
8	Sankofa	Ghana
9	Sankofa	Burkina Faso

Same title repeated

Different countries listed separately

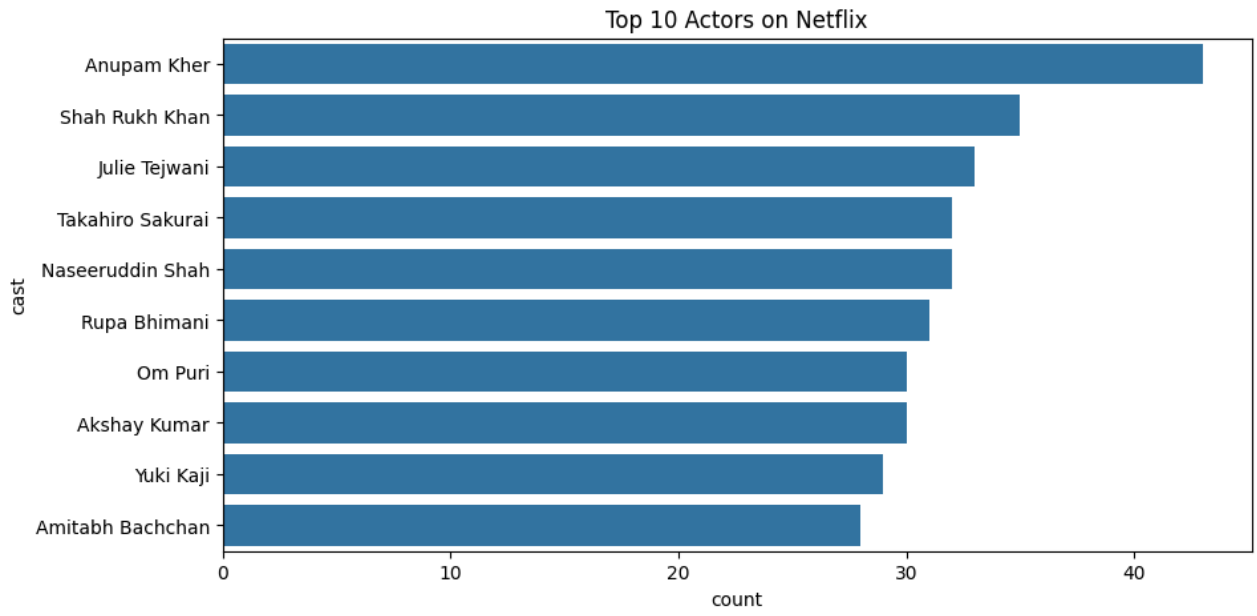
Countplot of top 10 countries

```
plt.figure(figsize=(10,5))
sns.countplot(
    data=country_df,
    y='country',
    order=country_df['country'].value_counts().head(10).index
)
plt.title('Top 10 Countries by Content Count')
plt.show()
```



Top 10 Actors

```
plt.figure(figsize=(10,5))
sns.countplot(
    data=cast_df,
    y='cast',
    order=cast_df['cast'].value_counts().head(10).index
)
plt.title('Top 10 Actors on Netflix')
plt.show()
```

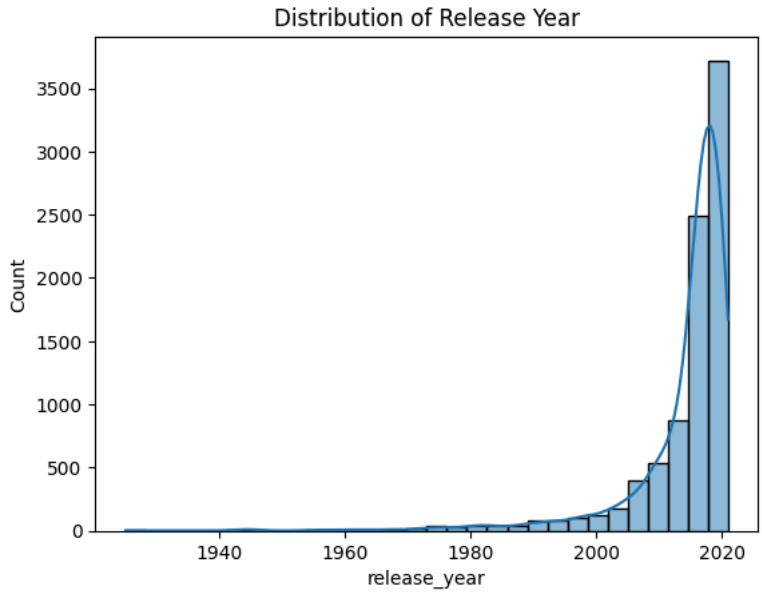


Pre-processing was performed by unnesting categorical columns such as cast, director, and country. Since these columns contain multiple values separated by commas, they were split and exploded into individual rows, reset indexs to enable accurate frequency-based and visual analysis.

4.1 UNIVARIATE ANALYSIS

A) Release Year

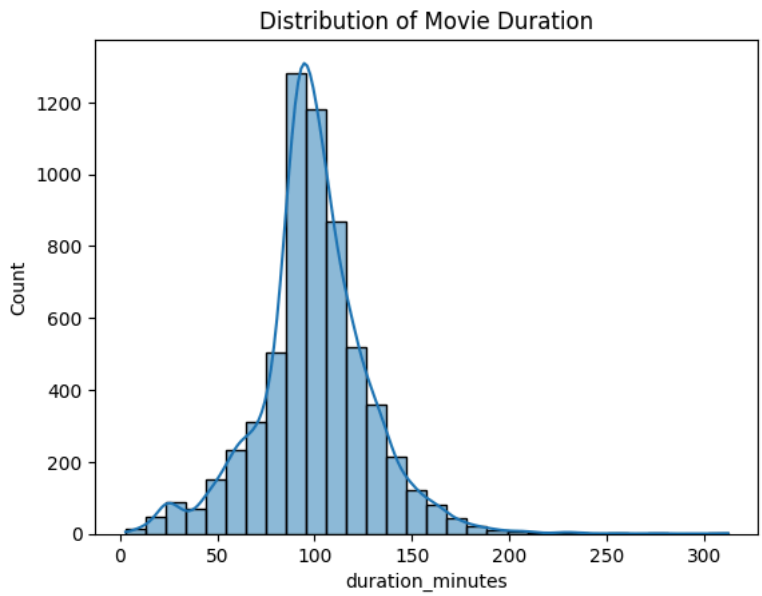
```
sns.histplot(data['release_year'], bins=30, kde=True)
plt.title('Distribution of Release Year')
plt.show()
```



Most content has been released after 2000, with a sharp increase after 2015

B) Movie Duration

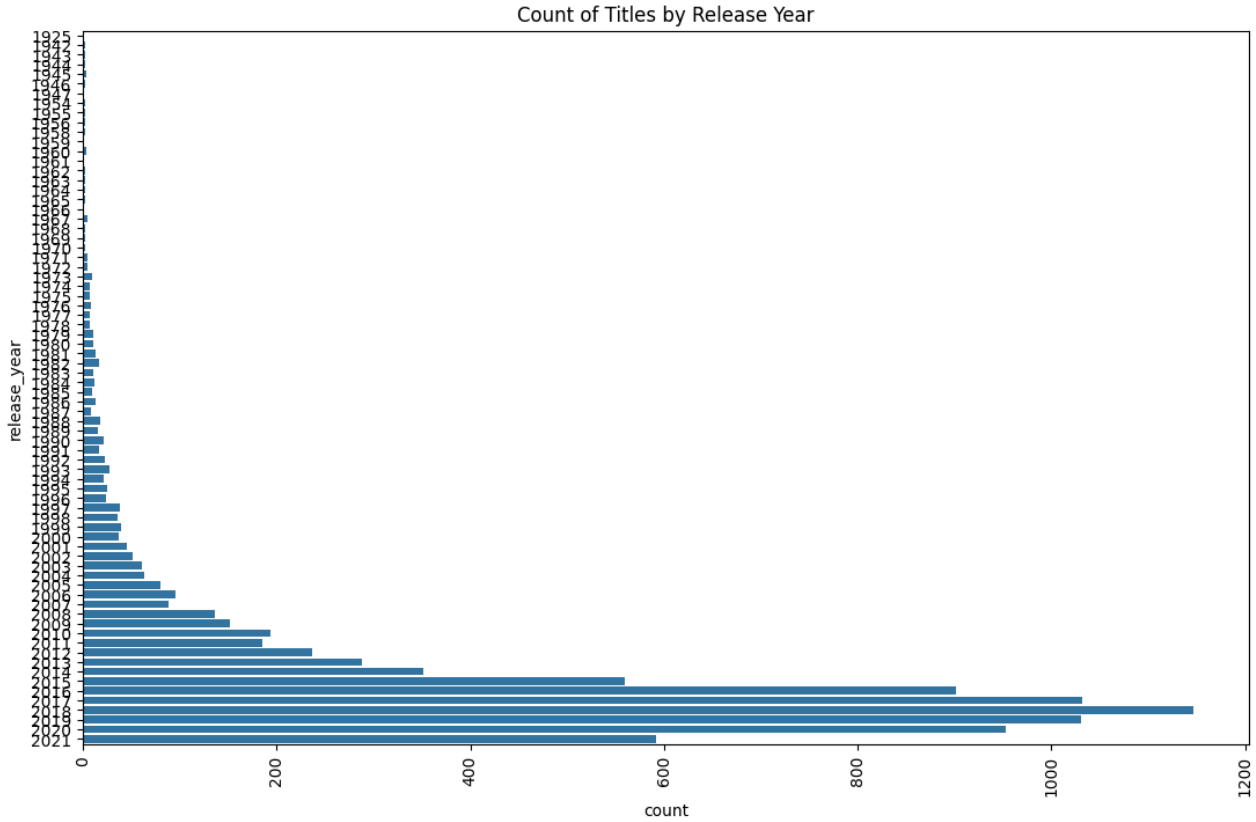
```
sns.histplot(movies_clean['duration_minutes'], bins=30, kde=True)
plt.title('Distribution of Movie Duration')
plt.show()
```



Most movies are between 80–120 minutes

C) Countplot (Discrete Numeric)

```
plt.figure(figsize=(13,8))
sns.countplot(y='release_year', data=data)
plt.xticks(rotation=90)
plt.title('Count of Titles by Release Year')
plt.show()
```

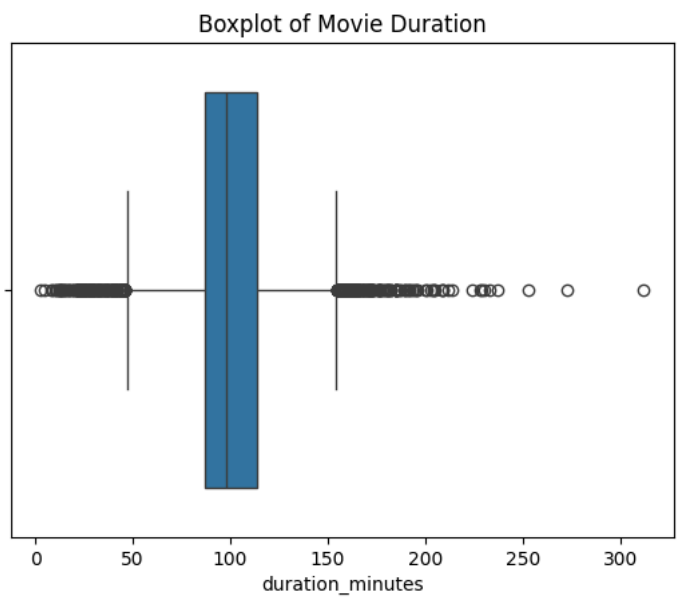


It shows from year 2000 onwards count of released years gradually increased till 2018.

4.2 BIVARIATE ANALYSIS (TWO VARIABLES)

Boxplot: Duration vs Type

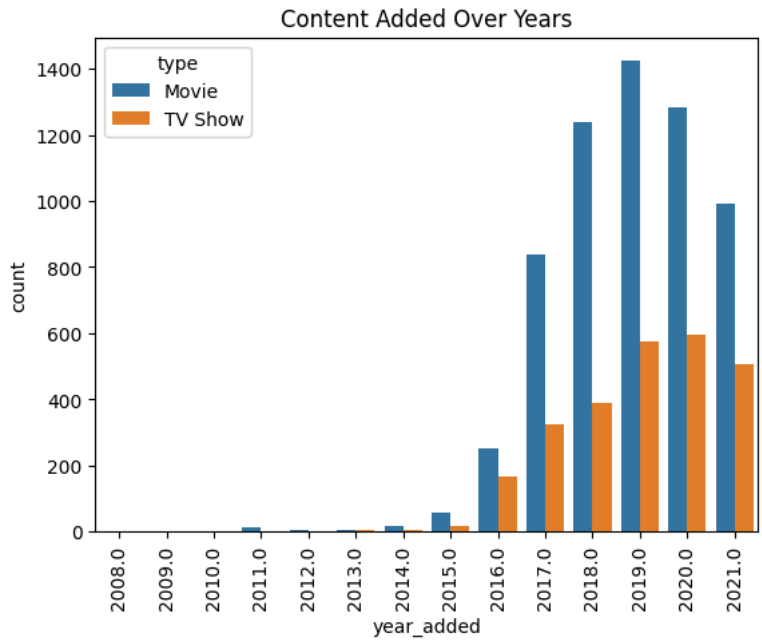
```
sns.boxplot(x=movies_clean['duration_minutes'])
plt.title('Boxplot of Movie Duration')
plt.show()
```



Shows spread, outliers

Movies vs TV Shows over time

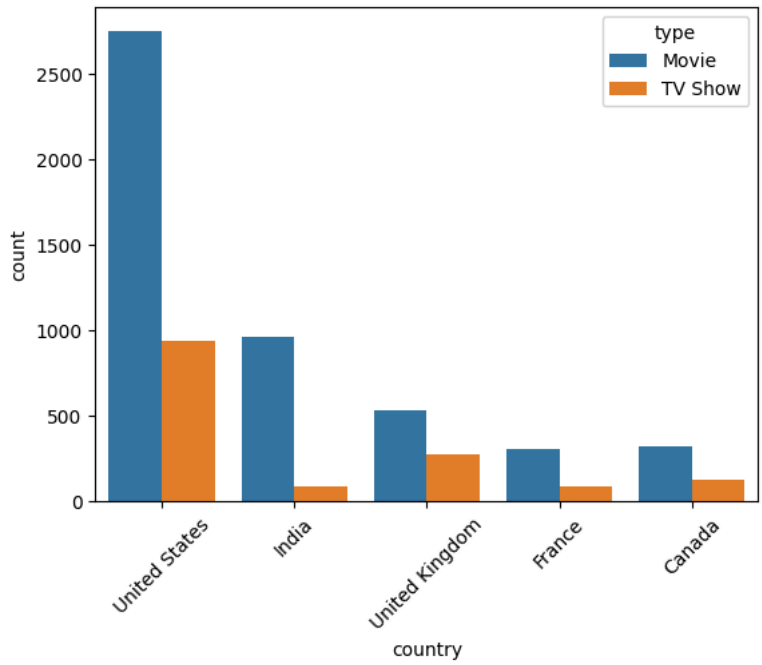
```
sns.countplot(data=data, x='year_added', hue='type')
plt.xticks(rotation=90)
plt.title('Content Added Over Years')
plt.show()
```



TV Shows increased significantly after 2016

Country vs Content Type

```
top_countries = country_df['country'].value_counts().head(5).index
sns.countplot(data=country_df[country_df['country'].isin(top_countries)],
              x='country', hue='type')
plt.xticks(rotation=45)
plt.show()
```



Explicit unnesting

5. **Missing Value & Outlier check (Treatment optional)**

Missing Value Check

```
data.isnull().sum()
```

	0
show_id	0
type	0
title	0
director	2634
cast	825
country	831
date_added	98
release_year	0
rating	4
duration	3
listed_in	0
description	0
year_added	98
month_added	98

dtype: int64

```
(data.isnull().mean() * 100).round(2)
```

	0
show_id	0.00
type	0.00
title	0.00
director	29.91
cast	9.37
country	9.44
date_added	1.11
release_year	0.00
rating	0.05
duration	0.03
listed_in	0.00
description	0.00
year_added	1.11
month_added	1.11

dtype: float64

This gives percentage missing values per column

Missing Value Analysis

The dataset contains missing values in multiple columns.

Columns such as director, cast, and country have a significant number of missing values.

Columns like date_added, rating, and duration have negligible missing values.

Missing values are expected in metadata fields, especially for older or regional content.

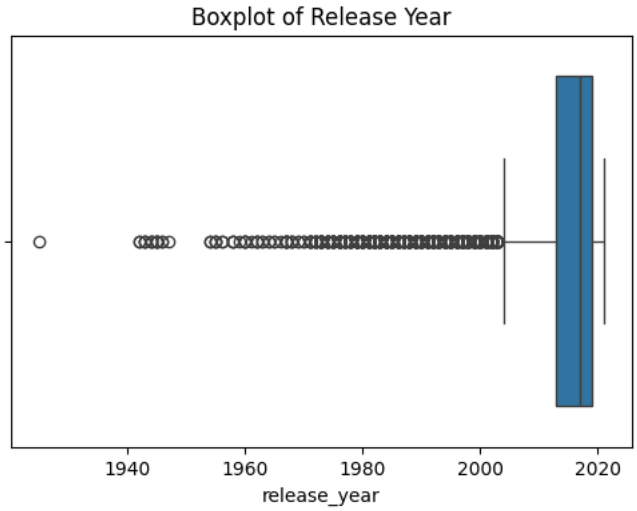
Outlier Detection

Numerical columns:

release_year

duration_minutes (movies)

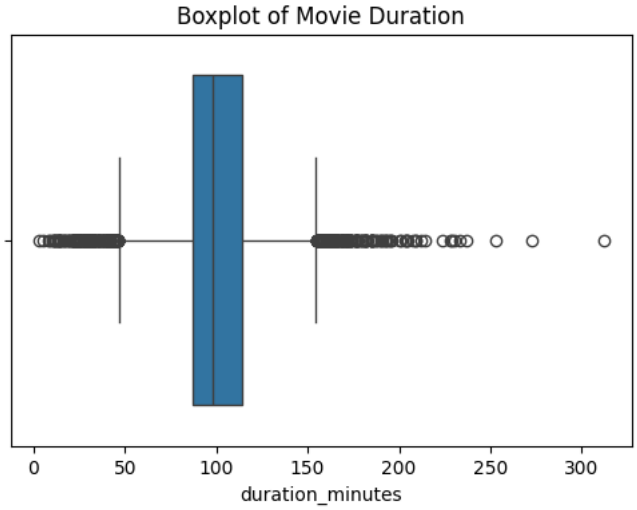
```
plt.figure(figsize=(6,4))
sns.boxplot(x=data['release_year'])
plt.title('Boxplot of Release Year')
plt.show()
```



Boxplot for Release Year

The release year does not show significant outliers, as the values fall within a reasonable historical range

```
plt.figure(figsize=(6,4))
sns.boxplot(x=movies_clean['duration_minutes'])
plt.title('Boxplot of Movie Duration')
plt.show()
```



Boxplot for Movie Duration

A few extreme movie durations are observed, which may represent documentaries or special editions. These outliers were retained as they represent valid content.

Missing Value and Outlier Analysis

The dataset contains missing values in columns such as director, cast, and country. These missing values are expected due to incomplete metadata for certain titles and do not significantly affect the overall analysis. Therefore, no imputation was performed.

Outlier analysis was conducted on numerical variables such as release year and movie duration. No significant anomalies were observed in the release year. A few outliers were present in movie duration; however, these values represent valid content and were retained.

6.Insights based on Non-Graphical and Visual Analysis

6.1 Comments on the Range of Attributes

The dataset contains content released over a wide time range, with release years spanning from the early 1940s to 2021. Movie durations range from short films of less than 60 minutes to long-form content exceeding 200 minutes. TV shows range from single-season series to multi-season shows with more than 10 seasons.

6.2 Distribution of Variables & Relationships

The distribution of content shows a strong skew towards movies compared to TV shows. In recent years, there is a noticeable increase in the number of TV shows added to the platform, indicating a shift towards episodic content. Content production is highly concentrated in a few countries, primarily the United States, followed by India and the United Kingdom.

6.3 Comments for Each Univariate & Bivariate Plot

Univariate Analysis – Type of Content

Movies constitute the majority of the content available on Netflix. However, TV shows, though fewer in number, have increased steadily over time, suggesting Netflix's growing focus on long-term viewer engagement.

Univariate Analysis – Movie Duration

Most movies have a duration between 80 and 120 minutes, indicating a preference for standard-length feature films. Very short and very long movies are comparatively rare.

Univariate Analysis – Country

Content production is dominated by a small number of countries, with the United States contributing the largest share. Emerging markets like India show significant representation, indicating Netflix's investment in regional content.

Bivariate Analysis – Content Type vs Year

Analysis of content type across release years reveals that TV shows have increased more rapidly than movies in recent years. This trend suggests a strategic shift towards binge-worthy series.

Bivariate Analysis – Country vs Content Volume

Countries with higher content volume tend to have diverse genres and both movie and TV show formats, highlighting Netflix's strategy of deep market penetration in key regions.

7. Business Insights - Should include patterns observed in the data along with what you can infer from it

1. Strong dominance of Movies, but rising focus on TV Shows

Movies form the majority of Netflix's content library.

However, in recent years, the number of TV Shows added per year has increased faster than movies.

This indicates a strategic shift by Netflix towards episodic and long-term engagement content, which helps retain subscribers for longer durations.

Inference: TV Shows are becoming a key growth driver for Netflix due to higher watch time and repeat engagement.

2. Content addition has accelerated after 2015

The number of titles added to Netflix has grown significantly after 2015.

Peak content addition is observed between 2017 and 2020.

This aligns with Netflix's global expansion phase and increased investment in original content.

Inference: Netflix scaled its content aggressively to support global market penetration and subscriber growth.

3. The United States is the largest content producer

The US contributes the highest number of movies and TV shows.

India, UK, Canada, Japan, and South Korea also show strong content presence.

Regional content from India and South Korea has grown noticeably in recent years.

Inference: Netflix relies on US content for volume but is increasingly investing in localized content to attract regional audiences.

4. Popular genres are Drama, Comedy, and International Content

Drama-related genres dominate across both Movies and TV Shows.

International Movies and TV Shows form a significant portion of the catalog.

Comedy and Documentaries also show consistent presence.

Inference: Netflix prioritizes emotion-driven and universally relatable genres that perform well across cultures.

5. TV-MA and TV-14 are the most common ratings

Most content is rated TV-MA (Mature Audience) and TV-14.

Family-friendly and kids' content forms a smaller share of the library.

Inference: Netflix primarily targets teen and adult audiences, focusing on mature storytelling rather than children-focused programming.

6. Content duration patterns differ by type

Movies typically range between 80–120 minutes, indicating standard feature-length productions.

TV Shows are mostly 1–3 seasons, suggesting Netflix experiments with limited-series formats.

Inference: Netflix prefers shorter TV series initially, likely to test audience response before committing to longer seasons.

7. A small set of directors and actors appear repeatedly

Certain directors and actors appear multiple times across different titles.

This indicates ongoing collaborations and trusted creative partnerships.

Inference: Netflix leverages successful creators and familiar faces to reduce risk and increase content acceptance.

8. Recommendations - Actionable items for business. No technical jargon. No complications. Simple action items that everyone can understand

1. Increase investment in TV Shows for subscriber retention

TV Shows encourage users to stay subscribed longer due to episodic viewing and multiple seasons.

Action: Produce more original TV Shows, especially limited-series formats (1–3 seasons), to maintain long-term viewer engagement.

2. Expand localized content in high-growth countries

Countries like India, South Korea, and Japan show increasing content presence and strong audience demand.

Action: Invest in region-specific stories, local actors, and native languages to attract and retain international subscribers.

3. Focus on high-performing genres

Drama, Comedy, and International content consistently dominate the Netflix catalog.

Action: Prioritize content development in these genres while experimenting within them (e.g., crime dramas, romantic comedies).

4. Maintain focus on mature and young-adult audiences

Most Netflix content is rated TV-MA and TV-14, indicating strong engagement from teenage and adult viewers.

Action: Continue producing bold, mature storytelling while gradually increasing family-friendly content to widen the audience base.

5. Strengthen partnerships with successful creators

Repeated appearances of certain directors and actors suggest proven creative success.

Action: Build long-term partnerships with high-performing directors and actors to ensure consistent content quality and audience trust.

6. Optimize content release timing

Content additions peak during certain months, suggesting strategic release planning.

Action: Schedule major content launches during high-viewership periods (festive seasons and holidays) to maximize reach.

7. Use data to test and scale content ideas

Short-duration TV Shows and limited seasons allow Netflix to test audience interest.

Action: Pilot new shows with limited episodes and scale up successful ones based on viewer response.

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